

INDEPENDENT ELECTRONICS PROJECT
Personal Design and Build — Portfolio Documentation

LM317 ADJUSTABLE POWER SUPPLY

Technical Design, Simulation and Experimental Validation Report

Samuel David Medina Contreras
Electronic Engineering Student

August 2026

Abstract

This report documents the design, simulation, construction, and experimental validation of an adjustable DC power supply based on the LM317 linear voltage regulator. The circuit converts 120 VAC mains power into a regulated, low-ripple DC output suitable for powering electronic laboratory prototypes. The design was first modeled in LTspice, then implemented on a soldered prototype board and enclosed in a protective case with an integrated digital voltmeter-ammeter for real-time monitoring. Experimental measurements — captured with a digital oscilloscope and a bench multimeter — confirm full-wave rectification at approximately twice the mains frequency, a low residual ripple of 160 mVpp at the regulated output, and a stable average output voltage of 5.90 V. The results validate the theoretical design and demonstrate a functional, reproducible power-supply module.

Index Terms

LM317, linear voltage regulator, full-wave rectifier, ripple voltage, DC power supply, LTspice, prototype board.

1. Introduction

Regulated DC power sources are a fundamental building block in electronic laboratories and prototyping environments. Most analog and digital circuits require a stable, low-noise supply voltage to operate correctly, and fluctuations or excessive ripple on the supply rail can introduce measurement errors, unwanted noise, or even damage to sensitive components.

In academic and hobbyist laboratories, adjustable linear power supplies remain a popular choice because of their simplicity, low cost, and predictable behavior compared to switching alternatives. Among the available linear regulators, the LM317 is widely used due to its adjustable output range, built-in thermal and current-limiting protection, and minimal external component count.

This project implements a complete AC-to-DC conversion chain — transformer, full-wave rectifier, capacitive filter, and LM317-based regulation stage — culminating in a bench-usable power supply with real-time voltage and current monitoring. The following sections describe the system design, present the LTspice simulation, document the hardware implementation, and report the experimental validation performed with an oscilloscope and a digital multimeter.

2. Objective

To design, simulate, build, and experimentally validate an adjustable DC power supply using the LM317 linear voltage regulator, verifying correct operation of the rectification, filtering, and regulation stages through bench measurements.

3. System Description

The power supply is organized into five functional stages, summarized in Table I and described in detail in the following subsections.

TABLE I
System Stages Overview

Stage	Description
Transformer	12 VAC step-down transformer, mains-isolated, feeding the rectifier bridge.
Rectifier	Full-wave bridge built with four 1N4007 diodes.
Filter Stage	Electrolytic capacitors (4700 μ F) providing bulk energy storage and ripple smoothing.
Regulation Stage	LM317 adjustable three-terminal linear regulator with feedback resistor network.
Monitoring Module	Digital voltmeter–ammeter display for real-time output monitoring.

3.1 Transformer

A 12 VAC step-down transformer isolates the circuit from the mains supply and steps the voltage down to a level suitable for rectification, providing the AC input to the bridge stage.

3.2 Rectifier

A full-wave bridge rectifier built from four 1N4007 diodes converts the 12 VAC input into a pulsating DC waveform, doubling the effective ripple frequency relative to the mains frequency (60 Hz $\rightarrow$ $\sim$ 120 Hz).

3.3 Filter Stage

Two 4700 μ F electrolytic capacitors smooth the rectified waveform by storing charge during the peaks and releasing it during the troughs, substantially reducing the ripple amplitude before regulation.

3.4 Regulation Stage

An LM317 adjustable three-terminal linear regulator, configured with a resistive feedback divider, sets and stabilizes the final DC output voltage, rejecting residual ripple and line variations.

3.5 Monitoring Module

A digital voltmeter-ammeter module is integrated at the output to provide real-time visual feedback of the delivered voltage and current, supporting bench use and troubleshooting.

4. LTspice Simulation

Prior to hardware implementation, the circuit was modeled and analyzed in LTspice to verify the expected behavior of the rectification, filtering, and regulation stages under nominal load conditions. The simulation confirmed the theoretical full-wave rectified waveform, the expected ripple reduction after filtering, and the target regulated output voltage, providing a baseline against which the experimental measurements in Section 6 are compared.

Note: the LTspice schematic capture is not included as a figure in this revision of the report; it can be added here (Fig. 1) alongside the corresponding simulated waveforms once exported from the simulation environment.

5. Hardware Implementation

The circuit was assembled on a soldered prototype (perforated) board, housing the rectifier bridge, filter capacitors, and LM317 regulation stage, and was subsequently mounted inside a transparent plastic enclosure for protection and portability. Figure 1 shows the internal assembly, including the transformer, rectifier wiring, filter capacitors, and the regulator board. The digital voltmeter-ammeter module and the output potentiometer are mounted on the enclosure wall for direct access during bench use.

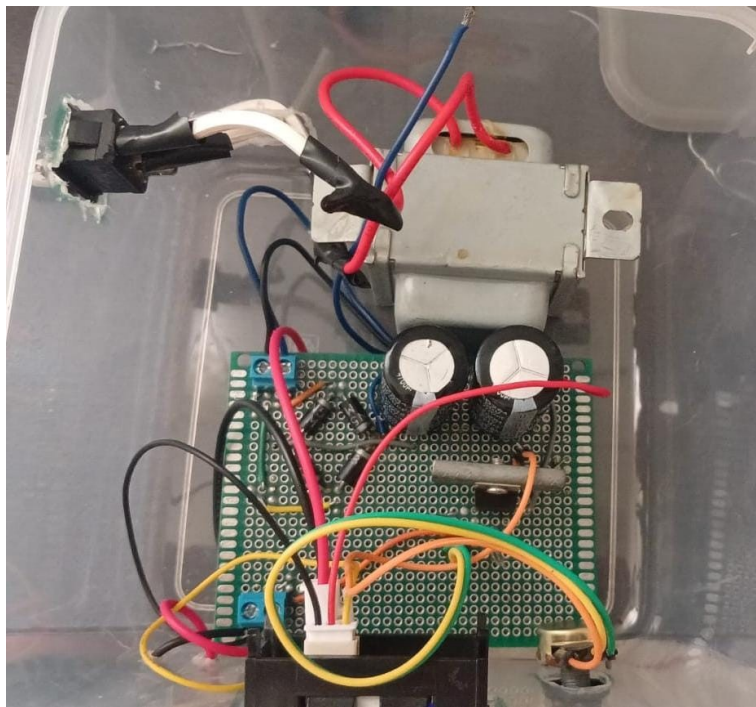


Fig. 1. Internal view of the assembled power supply, showing the transformer, rectifier wiring, filter capacitors, and regulator board mounted on the enclosure.

Figure 2 shows the completed enclosure during bench testing, with the integrated digital voltmeter-ammeter display reading the regulated output and a handheld multimeter used as an independent reference measurement.



Fig. 2. Completed power supply enclosure under bench test, with the onboard voltmeter-ammeter display and an independent multimeter reference measurement.

6. Experimental Validation

6.1 Full-Wave Rectification

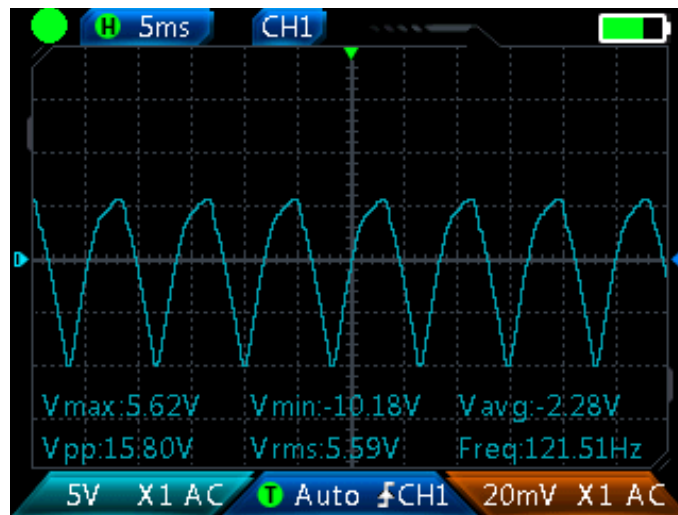


Fig. 3. Oscilloscope capture of the rectified output at the bridge, before filtering. Frequency ≈ 121.5 Hz, $V_{pp} \approx 15.8$ V.

The oscilloscope capture in Fig. 3 shows a full-wave rectified waveform with a measured frequency of approximately 121.5 Hz — close to twice the 60 Hz mains frequency — confirming correct operation of the bridge rectifier. The peak-to-peak amplitude of approximately 15.8 V is consistent with the expected voltage swing from the 12 VAC transformer secondary after diode conduction drops.

6.2 Ripple Measurement

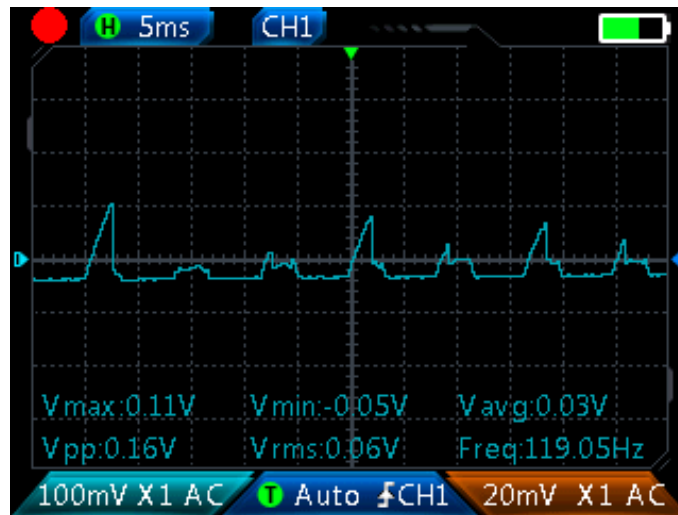


Fig. 4. AC-coupled oscilloscope capture at the regulated output, isolating the residual ripple. Frequency ≈ 119.05 Hz, ripple ≈ 160 mVpp.

To isolate the residual AC component riding on the regulated DC output, the oscilloscope was set to AC coupling. As shown in Fig. 4, the measured ripple was approximately 160 mVpp at a frequency of about 119 Hz, consistent with the charge/discharge cycle of the filter capacitors at twice the mains frequency. This low residual ripple confirms effective attenuation by the filtering and regulation stages.

6.3 Regulated Output Voltage

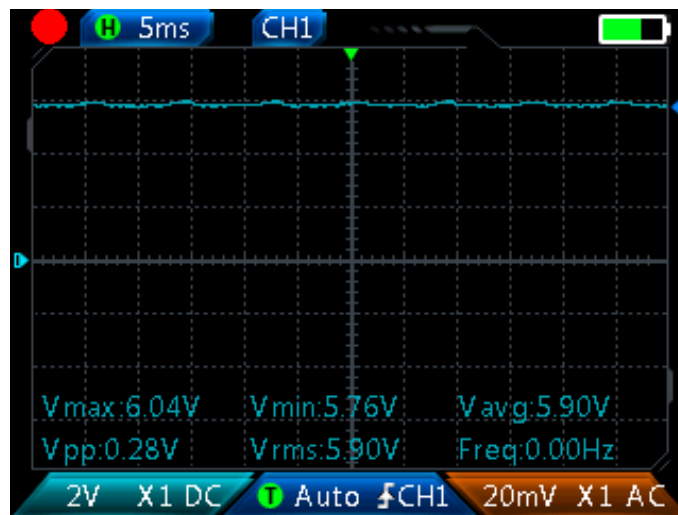


Fig. 5. DC-coupled oscilloscope capture of the regulated output. $V_{avg} \approx 5.90$ V, $V_{max} \approx 6.04$ V, $V_{min} \approx 5.76$ V.

The final regulated output was measured with DC coupling, yielding an average voltage of approximately 5.90 V, with a maximum of 6.04 V and a minimum of 5.76 V (Fig. 5). This narrow voltage band evidences stable, consistent regulation behavior consistent with the expected performance of the LM317 under the tested load.

7. Results Summary

TABLE II
Summary of Experimental Results

Parameter	Measured Value
Rectification Frequency	121.5 Hz
Rectified Ripple (raw, before regulation)	15.80 Vpp
Regulated Output Voltage (avg.)	5.90 V
Regulated Output Ripple	160 mVpp
Voltage Regulation	Stable

8. Discussion

The experimental measurements confirm proper operation of the rectifier, filtering, and regulation stages designed and simulated for this power supply. The full-wave rectified waveform exhibited the expected doubling of frequency relative to the mains, validating the bridge configuration. The 4700 μF filter capacitors reduced the raw rectified ripple substantially before the signal reached the regulation stage, and the LM317 further attenuated the residual ripple to approximately 160 mVpp at the output — a level well within the tolerance of most low-power electronic loads.

The regulated output remained stable at approximately 5.90 V across the measurement window, with a narrow deviation band of only 0.28 V peak-to-peak, demonstrating good line and load regulation for the tested configuration. Overall, the results are consistent with the theoretical LTspice simulation and confirm that the constructed prototype meets its design objective of delivering a stable, low-ripple DC output suitable for laboratory use.

9. Conclusions

1. The full-wave bridge rectifier operated correctly, producing an output frequency of approximately 121.5 Hz, consistent with twice the 60 Hz mains frequency.
2. The capacitive filtering stage significantly reduced the ripple amplitude prior to regulation, easing the demands placed on the LM317.
3. The LM317 maintained a stable regulated output close to 5.9 V, with residual ripple of only 160 mVpp, confirming suitable performance for low-power electronic applications.

10. Author

Samuel David Medina Contreras

Electronic Engineering Student — Independent Project